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Talking Environment on TikTok: Messages, Social Actors, and Engagement

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ABSTRACT

This study aims to provide an overview of the environmental discourse on TikTok by examining the message characteristics, social actors, and audience engagement with environment-related TikTok videos. We conducted a content analysis of 1515 TikTok posts in English and Spanish and found that: (1) videos about diverse environmental topics like vegan food, waste management, and pollution were more frequently posted and were relatively more engaging than climate change; (2) videos with lip-synced audio and narrative voices over a series of pictures received more engagement than original audio and vlog style videos respectively; (3) environmental messaging was largely ambiguous, but presenting environmental issues in a global context, sharing personal experiences, making emotional appeals, and revealing environmental beliefs explicitly predicted relatively greater engagement; (4) individual creators posted most content and received relatively more engagement than governmental and sustainability organizations. We provide recommendations for scientists and communicators based on these findings.

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Environmental TikTok; content analysis; planetary boundaries framework; multilingual analysis; social media

Today, much of the public discourse on issues demanding social change (e.g. climate change, the mental health crisis, and human rights) occurs on social media (Masip et al., 2019). This has decentralized the sources of information to include multiple social actors (e.g. individual influencers, NGOs, scientists, policymakers), enabling various such actors to participate in constructive dialogues and affect individual and collective decisions (Ghermandi et al., 2023; Ji et al., 2024; Nieto-Sandoval & Ferré-Pavia, 2024). Although the potential of social media in environmental communication appears promising, three crucial gaps in the scientific evidence base can be identified.

First, a predominance of studies on climate change communication has eclipsed efforts to understand people's awareness of other environmental processes affecting their lives (Comfort & Park, 2018; Hansen, 2011). Scientists have identified nine interlinked environmental phenomena that affect planetary health and quantified related boundaries which should not be crossed for humanity to thrive (O'Neill et al., 2018; Rockström et al., 2009). Alarmingly, six out of these nine boundaries were crossed by 2023, including those associated with chemical pollution and novel entities, biogeochemical flows, freshwater use, and land system (Richardson et al., 2023). While climate change is an important issue, concentrating the public discourse and scientific efforts only on climate change may prevent comprehensive public and political action with regards to other environmental problems (Barrios-O'Neill, 2021). Therefore, an overview of public discussions about environmental

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issues beyond climate change is warranted to be able to understand, and if needed, stimulate discussions on other issues like chemical pollution, wildlife destruction, and lack of access to freshwater, which affect billions of lives globally (UNEP, 2022).

Second, environmental communication literature on visual forms of media is limited (Comfort & Park, 2018; Pearce et al., 2019), and text-based social media like Twitter (now X) are researched more than platforms with audio-visual content (Ghermandi et al., 2023; Hansen, 2011; Pearce et al., 2019). However, video content has become increasingly popular, in part due to the rising prominence of TikTok. This social media app is used by over 1.8 billion people globally (i.e. nearly a quarter of all internet users) and comprised more active users than Twitter as of 2023 (Aslam, 2024). Moreover, TikTok is especially popular among young adults – 85% of users are aged under 35 years (Brown et al., 2022; McLean et al., 2023) – making it a compelling platform to examine how different message characteristics (e.g. environmental topics, audio-visual construction) and social actors (e.g. individuals, governments, NGOs) invite engagement from younger generations likely to experience the imminent dangerous consequences of environmental degradation. Such an understanding could help those intending to create effective pro-environmental messaging (e.g. influencers, policymakers, environmental NGOs) in adapting their social media posts to increase engagement with their content.

Third, much of social media research has focused on the environmental discourse of the English-speaking and is dominated by studies conducted among those in the Global North (Comfort & Park, 2018; Ghermandi et al., 2023). There have been calls for including environmental conversations in other languages to diversify our understanding of what constitutes the environment in the Global South – a region disproportionately affected by the consequences of environmental degradation (Acosta-Michlik et al., 2008; Blicharska et al., 2017). Social media content in languages like Spanish, which is spoken widely in the Global South (Montero et al., 2023) and represents the third largest user base on TikTok behind English and Mandarin (TikTok, 2024), can help highlight which environmental issues are prioritized in the region. Thus, in addition to English, we analysed TikTok videos in Spanish with the aim to shed a light on the environmental discourses in the Global South.

In this paper, we present a descriptive content analysis of environmental content on social media (i.e. TikTok) along with an examination of which message characteristics and social actors garner high engagement among the audience. Our study aims to advance scholarship on the aforementioned gaps by (a) providing an overview of the environmental discussions beyond climate change, (b) adding to the limited body of literature on audio-visual environmental communication, and (c) doing so by analysing content in two languages, i.e. English and Spanish. We also provide practical recommendations for various social actors like NGOs, sustainable brands, and governmental actors to create engaging social media content for fostering environmental awareness and behaviors.

Literature review

The field of environmental communication examines how individuals, groups, and organizations convey information about environmental issues to influence our understanding of the environment and our actions towards it (Comfort & Park, 2018; Hansen, 2011). It is both pragmatic – aiming to influence practices and policies – and constitutive, shaping how societies understand and engage with the natural world (Pezzullo & Cox, 2017). These pragmatic and constitutive aspects of environmental communication are increasingly conveyed on social media, via audio-visual content which was recently made popular on social media platforms, especially TikTok.

While audio-visual social media content is under studied in environmental communication, some research shows that videos could be preferred for environmental communication over text-based tools like pamphlets (Ahmad et al., 2015). Videos were also found to be more effective than text in communicating the scientific consensus about climate change (Goldberg et al., 2022), signaling the potential of utilizing TikTok videos to influence environmental awareness and actions. On TikTok and other social media, audience can respond to and engage with content in multiple

ways, ranging from more passive forms (e.g. viewing or liking a TikTok video) to more active forms (e.g. commenting on or sharing a TikTok) (Gainous et al., 2021; Guinaudeau et al., 2022; Muntinga et al., 2011). Active engagement with social media content was found to stimulate pro-environmental attitudes and behaviors (Debnath et al., 2022; Gibson et al., 2022; Robelia et al., 2011). Notably, for video content specifically, even passive engagement like viewing environmental videos on YouTube was linked to a strong positive effect on climate change awareness compared to other text-based social media (Gómez-Casillas & Gómez Márquez, 2023).

According to theories of classical communication, audience responses (i.e. engagement) is determined by how a message is constructed (i.e. message characteristics) and the source of the message (i.e. social actors) (Lasswell, 1948; McGuire et al., 2000). This is also true in the new media environments of the digital age, such as web platforms and social media apps (Metzger et al., 2003; Paek et al., 2013; Zheng et al., 2022). Indeed, different types of social media messaging and varied social actors influence sharing of social media content and can motivate further offline behaviors (Masip et al., 2019; Pavelle & Wilkinson, 2020). However, as explained above, our understanding of message and source effects for audio-visual environmental content on emerging social media like TikTok is lacking. Below, we summarize related findings from communication science (including environmental and health communication) to arrive at our research questions aimed at examining message characteristics and social actors of environmental videos on TikTok, and their effect on engagement (i.e. the number of plays, likes, comments, and shares).

Message characteristics

Message characteristics, like emphasizing emotions vs facts or including audio-visual features, may affect audience engagement (Corbett, 2012; Ha et al., 2019; Metzger et al., 2003). In this study, we examined the message characteristics of TikTok videos by adapting a typology from Shifman (2013) which is suited to the analysis of digital content, including social media posts. Specifically, we considered the *content*, *form*, and *stance* of environment related TikTok videos. Content comprises the main ideas communicated in a video, form refers to its physical construction, and stance depicts information about its purpose (Shifman, 2013).

Content

To understand the content of environmental TikTok videos, we analysed the primary topic, secondary topic(s), environmental ideology, and sentiment of the TikTok video. ***Primary topic*** is defined as the central environmental issue emphasized whereas *secondary topics* refer to the other peripheral environmental issues mentioned in the video. Despite the interconnectedness of environmental issues, studies of social media content have overlooked issues other than climate change, which was also found to be the most popular environmental topic of discussion on social media like Twitter (Dubey et al., 2024) as well as a major focus of environmental research on social media platforms to date (Hansen, 2011; Pearce et al., 2019).

Studies examining the environmental discourse on TikTok (e.g. Hautea et al., 2021; Nguyen, 2023) also seem to focus on climate change over other environmental threats outlined in the planetary boundaries framework, which identifies the nine critical environmental processes that maintain Earth's stability, such as biospheric integrity, ocean acidification, and geochemical cycles (Porkka et al., 2024; Richardson et al., 2023; Rockström et al., 2009). Communicating the framework in its entirety can help individuals grasp the urgency and interconnectedness of environmental issues, such as how water use, land-system change, and carbon emissions collectively impact planetary and human health (Häyhä et al., 2016). Furthermore, conveying the planetary boundaries can encourage a shift from an overwhelming focus on climate change toward recognizing the effects of human activity on various ecological systems, and fostering a more integrated understanding of ecological stewardship. To get a thorough picture of environmental issues that are discussed beyond, or along with climate change, we also examined the ***secondary topics*** in

TikTok videos. Such an examination of broader environmental topics on TikTok could inform which other topics are discussed by the public and their awareness about other natural phenomena.

Additionally, different topics can be communicated with a different underlying environmental ***ideology***, which is "a fully formed belief system concerning the natural world" (Corbett, 2012, p. 26). Ideologies comprise a spectrum of environmental beliefs ranging from an anthropocentric (human-centred) to an ecocentric (nature-centred) understanding of the natural environment (Corbett, 2012). Examining the environmental ideology underlying a TikTok video helps to understand the beliefs and values that motivate environmental behaviors (Bouman & Steg, 2019; Howell, 2013). Moreover, as ideologies develop in earlier in life (Chawla, 1999; Corbett, 2012), social media messaging may foster certain ideologies among the relatively younger users of TikTok at the expense of other ideologies. So, an overview of the prominent ideologies on TikTok videos is warranted to understand the environmental values these videos seek to, if any.

Lastly, content comprises ***sentiment***, referring to the overall opinion towards a topic (Pang et al., 2002). Pro-environment sentiment is conveyed when a message is intended to help protect the environment or prevent harm to it, whereas anti-environmental sentiment is expressed when the message aims to harm the environment or hinder its protection. Neutral sentiment occurs when no opinion towards the environment is expressed whereas ambiguous sentiment comprises both pro- and anti- environmental themes (Lange & Dewitte, 2019). Findings regarding public sentiment toward climate change on social media are mixed (Grouverman et al., 2018; Williams et al., 2015), but little is known about sentiments associated with environmental issues like biodiversity loss and freshwater access. We do know that environmental content with a negative sentiment is shared more widely on social media (Barrios-O'Neill, 2021), which implies that if negative messages about environmental issues other than climate change are also shared widely on TikTok, it could undermine the efforts to promote environmental action.

In sum, the content of an environmental TikTok video (i.e. its primary and secondary topics, environmental ideology, and sentiment) provide an understanding of the main ideas expressed in the TikTok video, which are important to examine as they may influence engagement differently. Next to the content itself, the way a message is presented influences engagement, especially on video-based platforms like TikTok.

Form

Form, or the "physical incarnation of a message, perceived through our senses" (Shifman, 2013, p. 367) comprises the *audio* and *video features* as well as the *genre* of the TikTok video. Compared to other platforms, TikTok's background ***audio features*** make it unique because even amateur users can easily create new videos by using already available music templates to generate content and relate to others (Vizcaíno-Verdú & Abidin, 2022). Similarly, individuals can easily modify the ***video characteristics*** on TikTok by editing original videos, adding video clips from the internet, and even collaborating with other video creators by providing their own video commentary (Kaye et al., 2022). Finally, ***genre*** refers to a broader categorization based on the style in which a video is created (Welbourne & Grant, 2016). For example, vlogs involve creators directly talking into the camera while presentations involve creators documenting an event without interacting with the camera themselves. There has been little research on employing audio features like sounds and music to engage the public in the environmental discourse (Boykoff & Beth Osnes, 2020), often restricted to studying the role of music in climate change messaging without measuring its effect on the audiences (Prior, 2022). Moreover, it is not yet known how video characteristics and genre are utilized for environmental communication on TikTok and which elements make a video particularly engaging. One recent study of viral TikTok videos (unrelated to the environment) found that text on videos facilitated its popularity rather than impede it (Ling et al., 2022), but a comprehensive examination of audio-visual characteristics is warranted to understand the full breadth of features that can make

environmental messaging on TikTok engaging. Another message characteristic that could impact the level of engagement is the stance, which is described next.

Stance

Stance refers to the communicative properties of a video and comprises *participation structures, keying,* and *communicative functions* (Shifman, 2013). ***Participation structures*** as adapted to environment related TikTok videos involve describing whose experiences are shared along with the context in which environmental topics are presented. On TikTok specifically, social actors tend to present climate change in both a local and global context, and relay more personal experiences rather than those of others (Nguyen, 2023), but it is unclear how such participation structures influence engagement. Previous research demonstrates that both local and global context offer distinct advantages under specific circumstances, but these effects have not been tested for social media like TikTok. As such, providing a local context can affect behavior by reducing the experienced psychological distance with environmental issues (Keller et al., 2022), whereas a global context could increase pro-social feelings towards those most vulnerable to the effects of the environmental crisis (Eijsink, 2022). These participation structures in environmental TikTok videos are important to investigate as they may affect construal of environmental issues as well as engagement with them.

Another stance related feature that might influence engagement with environmental posts is ***keying***, which affects the way a topic is perceived by the audiences (Blum-Kulka et al., 2004; Shifman, 2013). Keying refers to the tone of the video, such as humorous, sarcastic, sincere, etc. Research has found mixed results regarding the (positive and negative) responses to humorous video messages (Shriver-Rice et al., 2021), with some evidence that humorous messages are shared widely online (Hsieh et al., 2012). But engagement with messages containing sincere, sarcastic, or ironic tones is yet to be examined, and is crucial as emotions play an important role in how we think of and act towards the environment (Ettinger et al., 2021; Peters et al., 2022).

The final feature of stance, ***communicative function*** refers to the basic function of the video. For instance, TikTok videos can aim to provide information (referential function), evoke an emotional response (emotive function), persuade for action (conative function) among others. Previous findings regarding the effects of messages with different communicative functions are mixed. There is some evidence to suggest that emotional appeals were more effective than informational appeals in persuading indifferent audiences to adopt environmental behaviors (Hoewe & Ahern, 2017) and were found to positively influence individuals' attitudes towards water conservation (Bakar et al., 2023). However, when it comes to social media engagement, informational messages received more likes than emotional or infotainment-based appeals (Bakar et al., 2023; Davis et al., 2020), which were shared widely in comparison (Nikolinakou & King, 2018; Tellis et al., 2019). Finally, not much is known about the effects of conative appeals, with the exception of a study among young adults showing that the intention to share short-form videos about sea level rise was greater when the video contained a conative (action-oriented) aspect to it than when no follow-up action was suggested (Shriver-Rice et al., 2021). Given these mixed findings, it is difficult to conclude how the various message characteristics affect engagement. Hence, we ask:

RQ 1. What are the message characteristics of environment-related videos on TikTok?

RQ 2. How do different types of message characteristics of environment-related videos affect audience engagement (i.e. plays, likes, comments, and shares)?

Social actors

Public discourse involves many social actors (i.e. opinion leaders and sources) who shape our collective understanding of societal issues and influence public opinion and awareness. Online environmental discussions involve various social actors – including but not limited to individuals, influencers, journalists and media organizations, scientists, policymakers, as well as governmental,

non-governmental, and corporate organizations (Huber et al., 2022; Newman, 2017; Uldam, 2022). For instance, Newman (2017) found that following the release of Working Group I Summary for Policymakers in Intergovernmental Panel on Climate Change's (IPCC) Fifth Assessment Report, Twitter discussions were initially dominated by individual users, followed by media, activist, and governmental organizations – thereby demonstrating the multiplicity of social actors that exist on social media. A similar overview of environmental social actors on TikTok will clarify who drives the environmental discussion on this platform, with its unique demographics comprising mostly teens and young adults.

Moreover, different social actors may affect engagement with environmental messages differently. In one study exploring climate change on Twitter, content from news media organizations like newspapers and public broadcasting companies was found to be shared most often, followed by non-professional blogs and content produced by non-governmental organizations (Veltri & Atanasova, 2017). In a YouTube study of audience responses to information regarding threats of climate change in the U.S., messages attributed to climate scientists had a null or negative persuasive impact compared to messages attributed to Republican politicians and military leaders (Bolsen et al., 2019). Therefore, messages crafted to encourage pro-environmental actions may only be effectively engaging when issued via the right social actor. Hence, we ask the following research questions:

RQ 3. Who are the most prominent social actors creating environment-related videos on TikTok?

RQ 4. How does audience engagement differ among various social actors creating environment-related videos?

Contrasting English and Spanish language TikTok

Environmental communication researchers have called for more studies examining online environmental discourses beyond the English language, discussions in regions beyond Global North, and on social media platforms other than Twitter (Blicharska et al., 2017; Comfort & Park, 2018; Pearce et al., 2019; Tandon, 2021). These calls to action are based on the gaps that exist in our understanding of the environmental issues pertinent to the Global South – a region experiencing the consequences of environmental destruction disproportionately compared to the Global North. Examining non-English language discourses, especially those prominent in the Global South (e.g. Spanish) will enable insights into how a large part of the world discusses and prioritizes environmental issues. To this end, we will examine the role of language in the relationships between audience engagement and message characteristics.

RQ 5. To what extent does language (English vs. Spanish) influence audience engagement across types of message characteristics in environment-related videos on TikTok?

Methods

This study was approved by the Ethics Review Board of the Faculty of Social and Behavioural Sciences at the University of Amsterdam. The research questions[1] and analysis plan were preregistered on the Open Science Framework. Study materials including the content analysis codebook are uploaded on the OSF project page https://osf.io/bewr4/. The data for this study were retrieved using a third party TikTok API (TikAPI.io), implemented using Python. Analysis was conducted on R (R Core Team, 2021).

Data collection

Retrieving environment related TikTok videos

A mix of purposive and maximum variation sampling was employed to ensure that videos covering a diverse range of environmental topics were selected for this study.[2] Because videos on TikTok are

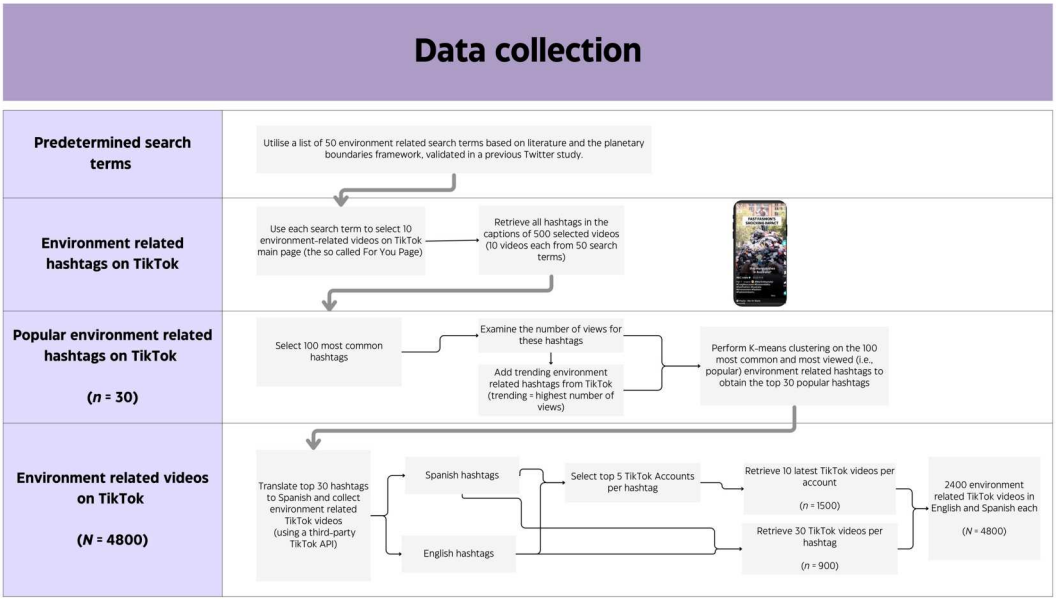


Figure 1. Retrieving popular environmental content on TikTok.

categorized and searched for using hashtags, we started by determining popular hashtags associated with various environmental search terms. By popular, we mean that these hashtags occurred most frequently in TikTok captions and were associated with the greatest number of plays (i.e. views). Figure 1 shows the steps of selecting popular hashtags ($n = 30$), followed by a description of how they were used to collect environment related videos on TikTok. In all, we retrieved 4800 videos in English ($n = 2400$) and Spanish ($n = 2400$) languages.

Step 1: retrieving environment-related hashtags. We used a pre-determined list of search terms ($n = 50$) to access popular environmental hashtags. This list was developed based on environmental research literature (Gaytan Camarillo et al., 2021; Veltri & Atanasova, 2017) and the planetary boundaries framework (Rockström et al., 2009) and validated for accessing environmental tweets for a previous study (Dubey et al., 2024). For each search term, we extracted all hashtags from the top 10 relevant (i.e. environment-related) videos on TikTok's For You page, where users can search for and view (recommended) content.

Step 2: selecting popular hashtags. After obtaining all hashtags related to the search terms, our next step was to select the most popular environmental hashtags based on frequency of occurrence as well as associated number of plays. First, the top 100 most frequently occurring environment-related hashtags were selected and their number of plays were examined. This list of hashtags was supplemented with results from trending-hashtag metrics provided by TikTok to include any remaining popular (i.e. trending) environmental hashtags that were missed. Second, we performed k-means clustering on frequency of occurrence and number of view counts per hashtag to establish the cut-off point for selecting the 30 most popular environment related hashtags with highest view counts. For obtaining videos in Spanish, two native Spanish speakers (back)translated this list of hashtags to Spanish and one native speaker verified the translations independently. These hashtags were then used to retrieve both top videos and top user accounts as described in the next step.

Step 3: retrieving environment-related videos. For English language content, we retrieved the top 30 videos and most recent 10 videos from top 5 creator accounts based on our list of hashtags. That is, we retrieved 900 videos (30 videos per hashtag) in each language directly from the API based on the content and retrieved 1500 videos from top creator accounts (10 videos for 150 creators; 5 creators per hashtag). We applied this approach to be able to include both content recommended by the platform and content from top environmental content creators.

Retrieving video information

For the retrieved videos, we also accessed publicly available metadata, including captions and engagement metrics (i.e. number of plays, likes, comments, and shares). To be able to categorize various user accounts as social actors, we obtained the profile information of the accounts that posted these videos. The profile information in Spanish was translated into English with the TextBlob library in Python (Loria, 2018) and manually verified by a native-Spanish speaker.

Data processing

Previous studies of environmental TikTok find that presence of climate change or sustainability related hashtags in captions did not necessarily mean that the content of the video is related to the environment (Hautea et al., 2021). As we retrieved TikTok videos based on varied environment related search terms and hashtags, it is likely that some videos were still irrelevant and did not relate to the environment. To remove irrelevant videos while retaining the diversity of environmental topics, we processed the collected data for further analysis by selecting what we considered environmental content, i.e. videos depicting the "interrelationships between people and the natural environment as well as videos depicting nature and natural phenomena" (*Aims and Scope – Journal of Environmental Psychology*, 2023). The environmental issues in the selected TikTok videos "include but are not limited to debates over climate change, natural resources, sustainability, conservation, wildlife, ecosystems, water, environmental health, food and agriculture, energy, and emerging technologies" (Environmental Communication Aims and Scope, 2023).

Moreover, duplicate videos posted by the same account were removed, but the same videos posted from different accounts were not removed as engagement might vary across accounts. We also included duets (i.e. picture-in-picture reaction videos) and stitches (i.e. video commentary on a video by another creator) in addition to original videos, as these features allow users to collaborate and interact with others and can be considered unique videos themselves. Table A1 in the appendix provides details of excluded videos in both English and Spanish.

Data analysis

We developed a codebook for conducting content analysis of environment related videos on TikTok (see supplemental materials). Based on the theoretical framework, we coded message characteristics including the content, form, and stance of the TikTok videos. The codebook was developed by analysing English language videos and validated using both English and Spanish videos. In the first step of codebook development, the first author began by inductively coding 102 videos (a 10% random subsample) to devise the categories for coding. Specifically, codes enumerating the primary and secondary environmental topics were devised and pre-existing typologies for other message characteristics were applied to the subsample to ensure that they were sufficient to describe the videos. This first version of the codebook was then deductively tested by all authors by iteratively coding the same sets of videos ($N = 30$ across three rounds) and disagreements were resolved in discussion.

The ensuing second version of the codebook was used to test its fit for analysing Spanish videos and to train the coders, resulting in two more revisions. As preregistered, we calculated the number of units to be analysed for reliability calculations based on the recommendation by Riffe et al. (2019,

p. 116), which indicated that the minimum number of videos to be analysed was 92. The fourth and final version was validated in multiple rounds of coding until sufficient inter-coder reliability was achieved separately for English and Spanish coders (*Krippendorff's* $\alpha = .83$ and .71 respectively; Krippendorff, 2004).

The results of the study are based on the remaining sample of 1515 TikTok videos in English ($n = 699$) and Spanish ($n = 816$), which were analysed by a team of three native Spanish speakers and two English speakers with full professional proficiency. As preregistered, this sample size was also sufficient to perform regression analysis to predict the level of engagement based on message characteristics and social actors. Specifically, the minimum required sample size was calculated to be 678 videos for the smallest detectable effect (i.e. exp $\beta 1 = 1.1$) with a power of 80% and significance level, $\alpha = .05$ in G* Power (Faul et al., 2007, 2009).

Results

The average duration of the videos was 43.97 s ($SD = 41.14$), with the longest TikTok being 6 min and 48 s long. Table 1 provides information about the engagement metrics of the videos, indicating that the videos received more passive engagement (number of plays and likes) than active engagement (number of comments and shares). The asymmetry between mean and median also shows that engagement wasn't distributed equally among the videos, i.e. a small number of videos received extremely high engagement compared to most other videos.

Message characteristics

To answer the first research question concerning message characteristics of environment related TikTok videos **(RQ 1)**, the content, form, and stance of the videos are described below.

Content

When looking at the content of the TikTok videos, biodiversity (i.e. flora, fauna, nature scenes, conservation, destruction, connection with nature/animals, etc.) was the most popular ***primary topic***, featuring in 21.82% of all videos. It was followed by videos related to sustainability (19.73% of the videos; e.g. tips for eco-friendly replacements, advertisements for sustainable brands). Notably, climate change was the prominent feature in only 7.06% of the videos, meaning that other topics took precedence over it. Waste management was the most popular ***secondary topic***, discussed in nearly a quarter (24.84%) of all videos. Figure 2 shows the prevalence of primary environment related topics. Table B2 in supplemental materials contains the frequency of secondary topics.

The ***environmental ideology*** was not explicitly stated by the creators in 92.68% of videos ($n = 1393$). Therefore, in addition to explicit mentions of ideologies, we also coded the implicit indicators as specified in the codebook. Anthropocentric ideologies were expressed to a much greater extent than ecocentric ideologies (Figure 3). Conservationism – the anthropocentric ideology that there should be some restraint in environmental resource use but only to preserve those resources for future human use seemed to underlie the messaging in 59.88% of all videos.

Table 1. Mean, median and standard deviation of engagement received by TikTok videos.

Engagement metric	*Mean*	*Median*	*SD*
Passive engagement			
Number of plays	3140227	75800	7913962.17
Number of likes	372900	4521	919047.29
Active engagement			
Number of comments	3061	70	11059.03
Number of shares	7475	114	28260.84

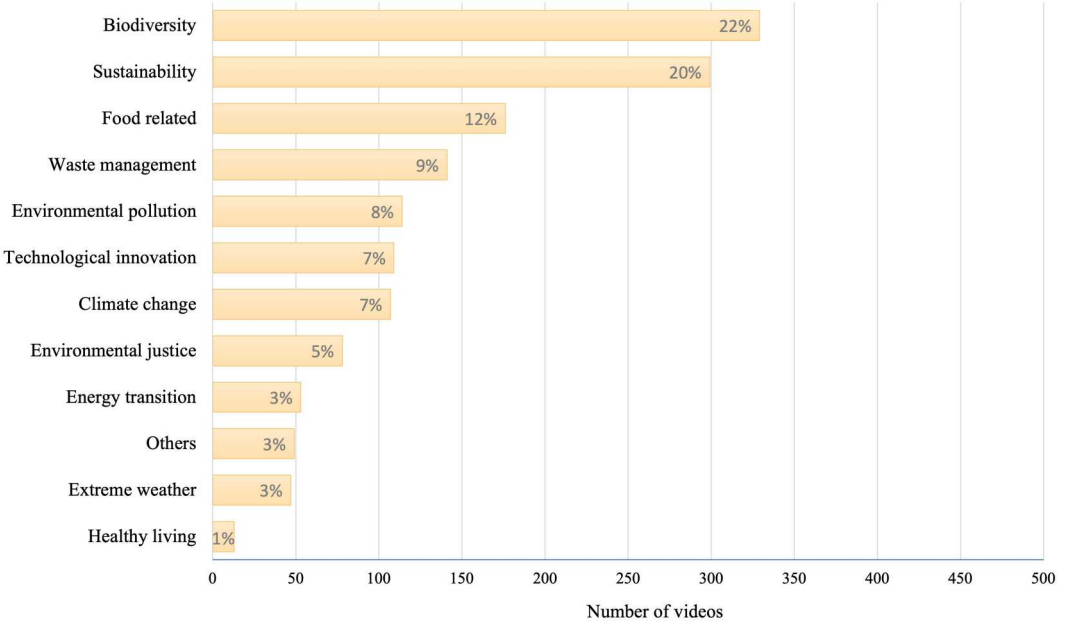


Figure 2. Primary topics in environment related TikTok videos.

With regards to the ***sentiment*** of TikTok videos (Figure 4), 55.64% of the videos were pro-environmental, propagating ways to help the environment and preventing harm to it. Surprisingly, the proportion of neutral messaging in videos was 42.84%, which means that many environmental-related videos conveyed neither positive nor negative messages towards the environment. Less than 1% of the videos contained anti-environmental (negative) or ambiguous (mixed) messages about the environment.

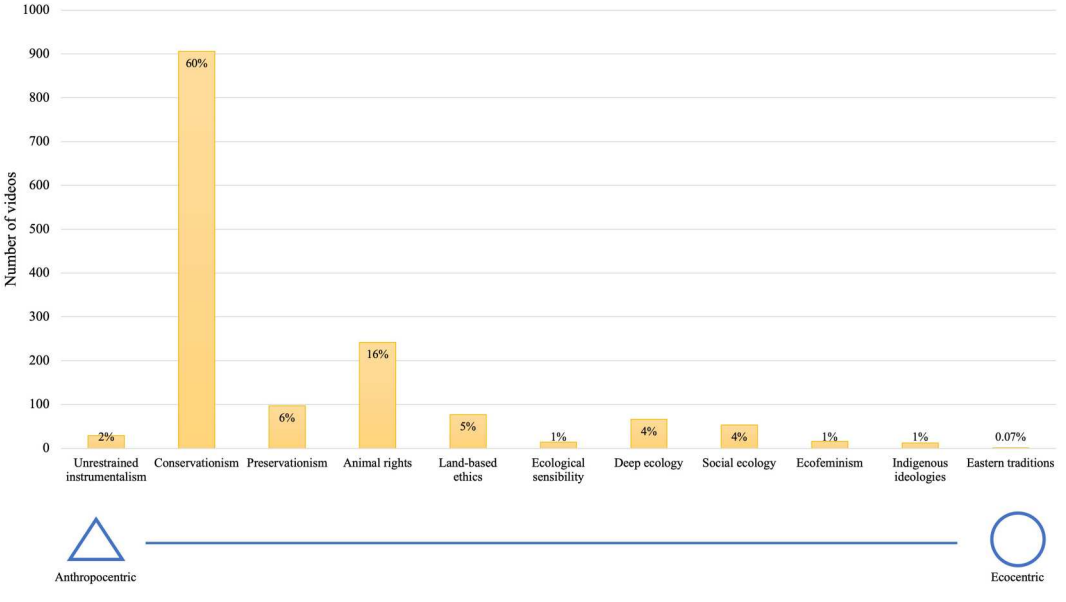


Figure 3. Environmental ideologies on TikTok.

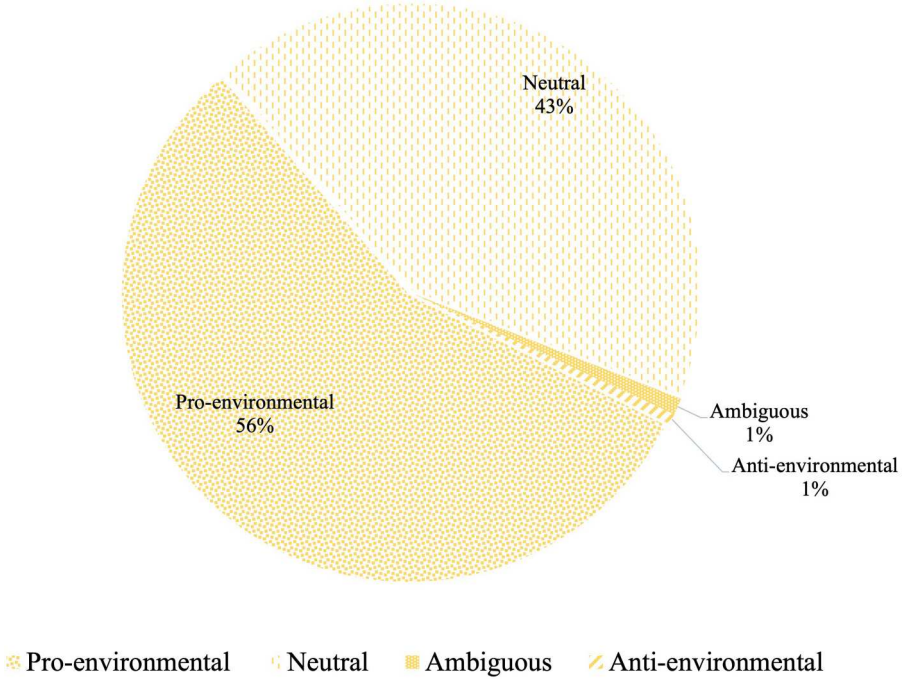


Figure 4. Environment related sentiment in TikTok videos.

Form

Table 2 summarizes the form of TikTok videos, comprising the audio characteristics, video features, and video genre. Overall, 32.87% of all TikTok videos contained original audio and 69.49% of videos were original videos. Among ***audio characteristics***, non-diegetic audio features were more prevalent (present in 41.67% of the videos) than diegetic audio features, which were present in 23.89% of the videos. In videos with original audio, 86.18% contained a human voice whereas the rest contained a computer-generated voice. Original videos were the most prominent ***video feature***

Table 2. Frequency of occurrence of audio characteristics, video features, and video genre on environment related TikTok.

Form	Frequency of occurrence	Percent
Audio characteristics		
Original	941	32.87%
Performance	45	1.57%
Diegetic	684	23.89%
Non-diegetic	1193	41.67%
Human voice	811	86.18%
Computer voice	130	13.81%
Video features		
Original video	1205	69.49%
Found video	403	23.24%
Animated video	28	1.61%
Timelapse video	16	0.92%
Interaction video	82	4.73%
Video genre		
Presentation	814	44.46%
Vlog	634	34.63%
Voice over pictures	183	9.99%
Text over pictures	113	6.17%
Hosted	59	3.22%
Interview	28	1.53%

(69.49%), followed by found videos from other sources (23.24%). With regards to the ***genre***, we found that most of the videos were filmed in either a presentation (44.46%) or a vlog (34.63%) style.

Stance

For stance, we described the participation structures, keying, and communicative function of the videos. For ***participation structures***, we identified the context in which the environmental topic was presented as well as whose experiences were depicted. We found that most creators (62.97%) were either unclear about the context or did not discuss environmental issues in an explicitly global or local context. Less than a third of the creators (28.51%) focused on their local context, for instance describing the plants in their backyard or recording a beach-clean up in their community. Environmental topics were discussed in a global context by 7.92% of the creators, such as while explaining global weather patterns or issues of environmental justice. In most videos (60.59%) it wasn't clear or explicit whose experiences (e.g. one's own or those of others) were being shared, followed by over a third of all videos (35.91%) in which creators narrated their own experiences. With regards to ***keying***, most videos were created to contain a serious (94.19% of all videos) tone, compared to mocking (2.44%), ironic (1.58%), funny (0.99%), or pretending (0.79%) tones.

Lastly, among ***communicative functions***, referential function was the most frequently utilized (in 60.99% of all videos; Figure 5), meaning that most videos aimed to provide information, whereas the metalingual communicative function aimed to define new concepts was the least popular (0.33% of all videos). Of all the videos employing the referential communicative function, only 2% aimed to correct a popular myth. Of those with conative function (i.e. videos containing calls to action; 25.15%), 93.44% aimed to encourage individual action compared to 6.56% that sought to encourage collective action (Figure 5).

Social actors

We identified 10 types of social actors across 726 unique creator accounts (Figure 6). Most videos were posted from individual accounts (e.g. influencer accounts; 61.57%) followed by aggregator accounts (i.e. accounts that repost other accounts' videos; 10.19%) and eco-friendly brand accounts (9.92%) that created content specifically to promote sustainable products. Sustainability organizations (6.34%) included NGOs, non-profits, and awareness organizations who used accounts to create awareness about the environment. Media actors (5.23%) included verified accounts of news

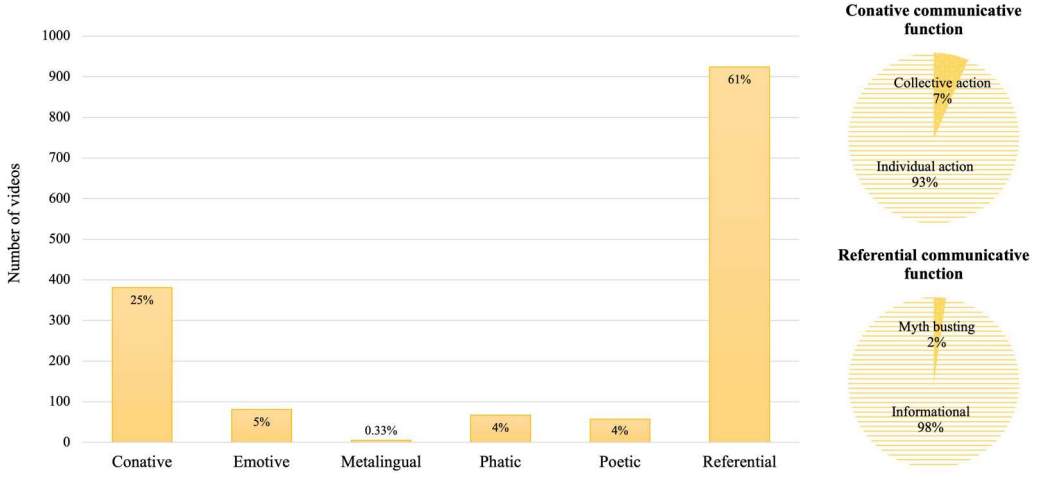


Figure 5. Communicative functions of environment related TikTok videos.

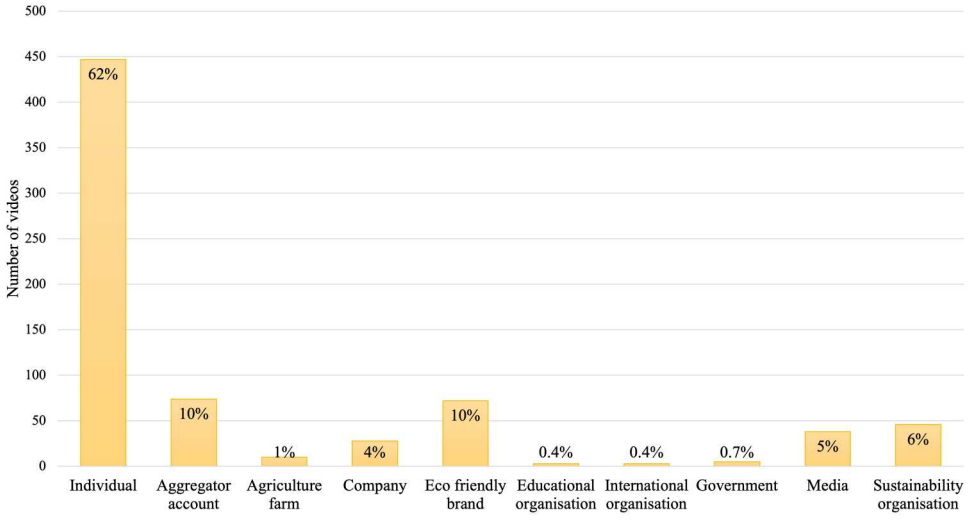


Figure 6. Social actors on environment related TikTok.

channels that posted environment-related news. Company actors (3.86%) included natural resources related industries such as those aimed at water purification and management. They were followed by agricultural farm accounts (1.38%) which posted content about farming tools and techniques, as well as technological developments in the farming sector. Content from governmental accounts (0.69%), international organizations (e.g. United Nations; 0.41%) and educational organizations (e.g. universities; 0.41%) was also found in our data, only to a lesser extent.

Engagement

To find out which message characteristics and social actors of a TikTok video predicted audience engagement in terms of number of plays, likes, comments, and shares, all predictors were entered in a negative binomial regression model as preregistered. When we were interested in specific comparisons based on previous research, we chose reference categories deliberately. This was the case for primary topic, allowing us to compare engagement received by videos on climate change to that of other primary topics. We also chose messages with unclear participation structures (context and experiences) as baseline for this reason. Referential categories for the ideology, audio characteristics, video features, genre, keying, and communicative functions were chosen based on the most frequently occurring characteristic. Exp(B) values were reported such that the expected engagement for a unit of change in the predictor variable could be calculated. Table A2 in the appendix provides a snapshot of all results, and the significant findings are discussed below in-text. Please see Tables B8–B11 in supplemental material for all coefficients and measures.

Engagement with message characteristics

Content. The results show that engagement differs greatly depending on the ***primary topic***. The topic of extreme weather event related videos for example evoked overall the least engagement. Compared to climate change, these videos received only 32% of the expected plays (exp(B) = 0.32, 95% CI [0.14, 0.17]), 39% of the expected likes (exp(B) = 0.39, 95% CI [0.17, 0.89]), 32% of the expected comments (exp(B) = 0.32, 95% CI [0.13, 0.75]), and 40% of the expected shares (exp(B) = 0.40, 95% CI [0.16, 0.99]). In contrast, videos related to plant-based food (e.g. vegan recipes) received greater engagement, than climate change, gaining 205% over the expected number

of plays (exp(*B*) = 2.05, 95% CI [1.06, 3.97]) and 247% over the expected number of likes (exp(*B*) = 2.47, 95% CI [1.26, 4.85]).

Since we coded both explicit and implicit indicators of ***ideology***, we first compared whether mentioning the environmental ideology explicitly was more engaging. When the underlying environmental ideology was explicit, videos tended to receive greater engagement in terms of number of likes (exp(*B*) = 1.54, 95% CI [0.94, 2.52]), comments (exp(*B*) = 2.05, 95% CI [1.21, 3.46]), and shares (exp(*B*) = 2.65, 95% CI [1.53, 4.58]), but not for the number of plays. For both explicit and implicit ideology combined, results showed that TikTok videos based on ecocentric ideologies (as opposed to anthropocentric ideologies) were more engaging to the audiences, with deep ecology based videos receiving more than double the plays (exp(*B*) = 2.28, 95% CI [1.21, 4.28]), likes (exp(*B*) = 2.69, 95% CI [1.41, 5.11]), comments (exp(*B*) = 2.38, 95% CI [1.21, 4.67]), and shares (exp(*B*) = 2.79, 95% CI [1.38, 5.64]) than expected as compared to conservationism.

Other ecocentric ideologies like ecofeminism and social ecology were associated with greater active engagement, as shown by the greater number of comments ($\exp(B)_{\text{ecofeminism}} = 9.85$, 95% CI [2.74, 35.43]; $\exp(B)_{\text{social ecology}} = 4.50$, 95% CI [1.95, 10.36]) and shares ($\exp(B)_{\text{ecofeminism}} = 6.16$, 95% CI [1.62, 23.44]; $\exp(B)_{\text{social ecology}} = 4.46$, 95% CI [1.86, 10.65]) than expected. Finally, compared with videos containing a neutral ***sentiment***, those with a pro-environmental sentiment received almost half the plays (exp(*B*) = 0.53, 95% CI [0.40, 0.71]) and likes (exp(*B*) = 0.53, 95% CI [0.39, 0.71]) than expected.

Form. Compared to original ***audio***, diegetic audio predicted a greater number of plays (exp(*B*) = 1.40, 95% CI [1.05, 1.88]) and likes (exp(*B*) = 1.42, 95% CI [1.05, 1.91]). The number of shares was greater for performance audio when compared to original audio (exp(*B*) = 2.94, 95% CI [1.21, 7.14]). ***Video features*** and ***video genre*** did not predict passive engagement (i.e. plays and likes) but were associated with active engagement (i.e. comments and shares). Specifically, compared to original video, found videos received more comments (exp(*B*) = 1.46, 95% CI [1.00, 2.13]) and were shared more often (exp(*B*) = 1.56, 95% CI [1.05, 2.31]). Similarly, for video genre, hosted videos (exp(*B*) = 2.12, 95% CI [1.10, 4.09]) and videos comprising voice over pictures (exp(*B*) = 1.89, 95% CI [1.25, 2.87]) received more comments compared to the presentation genre. Also compared to performance videos, vlog style videos were shared to a lesser extent (exp(*B*) = 0.66, 95% CI [0.46, 0.94]).

Stance. We found that the ***context*** of the videos affected engagement. Specifically, presenting environmental issues in a global context predicted a greater number of plays (exp(*B*) = 2.69, 95% CI [1.52, 4.74]), likes (exp(*B*) = 3.19, 95% CI [1.79, 5.70]), comments (exp(*B*) = 3.04, 95% CI [1.66, 5.58]), and shares (exp(*B*) = 2.63, 95% CI [1.40, 4.97]) when compared to videos with unclear context. For ***experiences***, we also found differences in engagement. Sharing personal experiences also garnered more engagement compared to when it was unclear whose experiences were shared (plays: exp(*B*) = 1.93, 95% CI [1.28, 2.92]; likes: exp(*B*) = 1.80, 95% CI [1.18, 2.75]; comments: exp(*B*) = 2.74, 95% CI [1.76, 4.26]; shares: exp(*B*) = 2.04, 95% CI [1.28, 3.23]). For ***keying***, videos where the social actor was pretending to be someone/something else received fewer plays (exp(*B*) = 0.16, 95% CI [0.04, 0.62]), likes (exp(*B*) = 0.13, 95% CI [0.03, 0.54]), and shares (exp(*B*) = 0.04, 95% CI [0.01, 0.17]), compared to serious videos in which the topic was sincerely represented. Finally, with regards to ***communicative functions***, videos with an emotional message were found to be more engaging compared to videos with an informational message, receiving more plays (exp(*B*) = 3.56, 95% CI [2.04, 6.23]), likes (exp(*B*) = 4.64, 95% CI [2.62, 8.21]), comments (exp(*B*) = 4.55, 95% CI [2.50, 8.27]), and shares (exp(*B*) = 3.65, 95% CI [1.95, 6.82]).

Engagement with social actors

The type of social actor who posted environmental videos seemingly predicted the level of engagement. Generally, compared to individual accounts, social actors such as aggregator accounts, media

organizations, sustainability organization etc. received less engagement (Table B12 in supplemental materials). Environment related posts from governmental accounts received fewer plays (exp(B) = 0.09, 95% CI [0.02, 0.39]), likes (exp(B) = 0.07, 95% CI [0.02, 0.34]), comments (exp(B) = 0.12, 95% CI [0.02, 0.58]), and shares (exp(B) = 0.15, 95% CI [0.03, 0.80]) than individual accounts. Similarly, videos posted from accounts of sustainability organizations garnered less engagement than expected, receiving fewer plays (exp(B) = 0.19, 95% CI [0.12, 0.28]), likes (exp(B) = 0.16, 95% CI [0.10, 0.24]), comments (exp(B) = 0.17, 95% CI [0.11, 0.26]), and shares (exp(B) = 0.12, 95% CI [0.07, 0.18]). For eco-friendly brands active engagement was also low, as shown by fewer number of comments (exp(B) = 0.37, 95% CI [0.25, 0.56]) and shares (exp(B) = 0.41, 95% CI [0.27, 0.63]) when compared to individual accounts.

Differences in engagement for English and Spanish videos

Language moderated the effect on level of engagement of several message characteristics, including the primary topic, the keying, and the communicative function. For instance, when looking at *primary topic* and holding all else constant, food-related videos in Spanish received 97% fewer plays (exp(B) = .03, 95% CI [.01, .13]), 91% fewer likes (exp(B) = .09, 95% CI [.02, .36]), and 84% fewer comments (exp(B) = .16, 95% CI [.04, .70]) than those in English. On the other hand, Spanish TikTok videos on the topic of environmental pollution garnered more active engagement, receiving more comments (exp(B) = 5.63, 95% CI [1.52, 20.92]) and more shares (exp(B) = 15.17, 95% CI [3.89, 59.13]) compared to videos in English language. The engagement for funny environmental TikTok videos also varied by language when everything else was constant. That is, videos in Spanish receiving nearly no plays (exp(B) = .00, 95% CI [.00, .01]), likes (exp(B) = .00, 95% CI [.00, .01]), comments (exp(B) = .00, 95% CI [.00, .04]), and shares (exp(B) = .00, 95% CI [.00, .01]) compared to those in English. Finally, engagement for TikTok videos with conative communicative function also differed by language, with videos in Spanish receiving over twice the number of plays (exp(B) = 2.78, 95% CI [1.54, 5.01]), likes (exp(B) = 2.92, 95% CI [1.61, 5.29]), and comments (exp(B) = 2.29, 95% CI [1.23, 4.28]), but not shares (p = .91).

Discussion

With the digital age ushering in new possibilities for environmental communication, social media could be a medium to foster environmental beliefs and behaviors. This study provides an overview of the environmental discourse on TikTok and adds to existing literature by addressing three previously highlighted research gaps (Comfort & Park, 2018; Ghermandi et al., 2023; Pearce et al., 2019). First, we add to the breadth of environmental issues examined in previous research by considering less researched, but equally relevant environmental phenomena than climate change, such as biodiversity loss and freshwater use. Second, we move beyond the already abundant literature on text-based platforms like Twitter by analysing audio-visual data to reflect the rising popularity of video-based social media like TikTok. Finally, we supplement the currently dominant English-language narratives in environmental research by also including videos in Spanish, as these may imply a different understanding of the environment than English, which tends to pivot towards highlighting only Global North perspectives. The findings of our study and their implications are discussed below.

Environmental discourse on TikTok is diverse beyond the topic of climate change

The data from this study indicate that on TikTok, climate change was only the seventh most popular environmental ***primary topic*** of discussion, preceded by topics related to biodiversity, sustainability, food, waste management, and environmental pollution. This diverse range of environmental topics suggests that unlike on Twitter (Chang et al., 2022; Dubey et al., 2024), issues beyond climate change are very much a part of the public discourse. This could be explained partly by platform features (like TikTok's) that aim to capture user attention by promoting novel stimuli (Lorenz-

Spreen et al., 2020). Specifically, climate change, which might be a somewhat saturated point of discussion on social media (Dubey et al., 2024; Hansen, 2011; Shreedhar, 2021) may not capture user attention as well as novel environmental information and hence posts about it may be promoted less frequently on the platform.

Topics other than climate change seem to be popular among the audiences on TikTok. Videos discussing plant-based diets, waste management, and environmental pollution received more engagement than climate change, indicating TikTok's potential to communicate about environmental issues of low public awareness. An implication for social actors like governmental and sustainability organizations who create pro-environmental messaging on social media is to diversify the range of environmental topics in their content, also discussing additional environmental problems alongside climate change.

Environmental messaging in TikTok videos is largely vague and unclear

We found that most environment related TikTok videos did not explicitly mention their environmental ***ideologies***. This could prevent especially newer audiences from understanding why they should adopt certain behaviors and whether they help or harm the environment. Environmental ideologies, or beliefs about the natural world, drive our environmental behaviors along with factors like available resources and values (Corbett, 2012; Steg, 2016). Making environmental ideologies explicit in TikTok videos, by providing reasons for caring about the environment (e.g. for preserving resources for future generations, or because nature has value outside of human use and we have a responsibility to not degrade it) could be more engaging, also because in our study, explicit ideologies predicted greater audience engagement on TikTok.

Many tweets used climate change related hashtags but did not express a positive or negative ***sentiment*** towards the environment. These findings are challenging because indistinctness in environmental messaging may prevent the audience from making connections between their ideological and value systems and environmental behaviors presented in a TikTok video, or their beliefs about the natural environment and lifestyle choices (Corbett, 2012; Novikau, 2016). For instance, many food-related videos espousing a plant-based diet did not express their ideological beliefs behind adopting such diets and remained neutral in their messages regarding the environment, making it difficult for the audience to relate their dietary habits with environmental impact. Hence, explicitly conveying underlying ideological beliefs and pro-environmental sentiment in social media messages might prove more effective in changing beliefs and behaviors.

Conveying global context, personal experiences, and emotions is engaging

While most videos did not explicitly provide a ***context*** to their discussion of environmental topics, presenting issues in a global context was found to be more engaging than when no context was provided. These results corroborate and advance findings in environmental research related to psychological distance, which highlight that local messages are only effective when they contain personally relevant and meaningful cues (Brügger, 2020; Eijsink, 2022). Moreover, global framing has been found to offer distinct advantages over local framing under some circumstances, such as the increases in perceived social benefits of taking environmental action (Dasandi et al., 2022; Spence et al., 2012). Hence, with social media audience tuning in from around the world, globally oriented messages like those aimed at saving the most vulnerable communities from the effects of environmental crises may strike a chord with the audiences by reducing the psychological distance to environmental phenomena.

The same was true for depiction of personal ***experiences*** compared to when it wasn't clear whose experiences were shared. This mirrors findings that personal narratives tend to have a greater effect on environmental behaviors and community mobilization than informational narratives (Barr & Gilg, 2006; Morris et al., 2019; Robinson, 2016). We also found that despite most TikTok videos

serving an informative ***communicative function***, emotional messages were overwhelmingly more engaging in comparison. It is possible that this high engagement is driven by the potential role emotions play in increasing virality of social media content (Botha & Reyneke, 2013; Macdonald et al., 2016; Nikolinakou & King, 2018). Moreover, emotional cues in user generated content on social media have been found to activate pro-environmental personal norms (i.e. degree of responsibility towards the environment) by making people aware of the consequences of their actions and by increasing their sense of accountability (Han et al., 2021). Future research could examine the impact of emotive messages on environmental beliefs, values, and actions to ascertain the extent to which such messages can persuade people to be environmentally friendly.

Tiktok's audio-visual features and language influence engagement differently

When it comes to the construction of the videos, ***audio characteristics*** seemed to predict engagement differently. For instance, videos where the creator was performing someone else's audio were shared more frequently than those with original audio. This could be explained by TikTok features which allow creators to use pre-existing audio templates for their videos, which means that viral (i.e. extremely popular) pre-existing templates can be used for creating messages that are shared more widely. Similarly, while ***video features*** did not predict passive engagement (number of plays and likes), found videos received greater active engagement in terms of more comments and shares than original videos. We found that videos in a more conversational ***genre*** received more comments than original videos, implying that videos containing (clips of) interviews or hosted programs could be more persuasive than when a single presenter is delivering a vlog style message. Content creators could also utilize infographics for provoking conversations on environmental topics, as such videos received more comments from the audience.

We also found differences in the relationships between message characteristics and engagement depending on the ***language*** of the TikTok video. For instance, food-related videos in Spanish received lower engagement than videos about climate change, whereas food-related videos in English received higher engagement. Environmental pollution emerged to be a more pressing environmental concern in Spanish TikTok videos, receiving higher active engagement from the audiences. This indicates the differences in discourses along geographical lines, highlighting environmental discussions from Global South countries including Chile, Argentina, and Colombia among others. The differences were not limited to prioritization of issues, as suggested by the finding that funny videos received more engagement than serious videos in English but not in Spanish. Hence, a humorous tone may not be universally effective in environmental messaging, implying that communicators should adapt their messages to suit their target audience on social media.

Individual social actors create most of the environmental content on TikTok

Most accounts that posted environment related TikTok videos were individual actors such as lifestyle influencers, activists, and hobbyists. Other social actors such as eco-friendly brands, international organizations (e.g. the United Nations) and media organizations were found to a lesser extent. These findings reflect a possible consequence of TikTok's features that "*lower the barrier to content creation and are adapted to the fast-paced information consumption of today's digital society …* " (Vizcaíno-Verdú & Abidin, 2022, p. 884), such that more individuals can create, distribute, and access video content on their areas of interest. This ease of use and popularity of TikTok – 9 out of every 10 users report accessing content on it every day (Kaye et al., 2022) – may make it a suitable communication channel to engage individuals for pro-environmental action.

We also found that environmental messaging being produced by individuals on TikTok received greater audience engagement than other social actors like sustainability organizations, media organizations, and governmental organizations. This is unorthodox, as studies show that messaging may be more effective when coming from favored organizations as compared to individuals

(Flanagin & Metzger, 2007). For instance, Kapoor et al. (2022) found that sustainability messages were more effective when posted on social media by an eco-friendly hotel than a social media influencer. One reason that individual accounts receive more engagement than organizational accounts could be that social actors who are similar to their audiences tend to be more persuasive than dissimilar social actors (Perloff, 2003). Similarly, the importance of perceived similarity may outweigh the value of perceived expertise among young people, who form the core demographic of TikTok users (Ahn et al., 2009; Paek et al., 2011).

While the predominance of individual content creators is welcome for increasing democratization of sources of environmental information, it has also resulted in concerns regarding the quality and veracity of this information (Basch et al., 2022; Hautea et al., 2021; Nieto-Sandoval & Ferré-Pavia, 2024). Our findings encourage collaborations between individuals and organizations on TikTok, a platform known for its collaborative features (e.g. duets and stitches), such that audiences can have access to quality information provided by a face they recognize or relate with, making it more likely that the environmental message is received well.

Limitations and future directions

Some limitations of our study warrant discussion such that they can be improved upon in future studies. First, while we used a wide range of search terms, we could still be missing important environmental search terms that are specific to social media platforms (e.g. #ecotok). We used the scientifically developed planetary boundaries framework to arrive at popular environmental hashtags, but being able to obtain popular hashtags from TikTok itself would ensure an analysis of the most topical environmental discussions. The TikTok API, released after this study was completed, could make it easier to conduct such analyses (Upton, 2023). Second, the findings of this study represent a snapshot of environmental discussions on TikTok, based on the data collected at a specific time point and at a certain geographical location. While the creation of a new user account for collecting the hashtags minimized the bias in sampling (compared to a pre-existing account), the algorithmic recommendation on TikTok shapes the content shown to users to a greater degree than other platforms (Guinaudeau et al., 2022), and likely influenced the selection of content for this study as well. Future research could use a similar methodology across different times and locations to determine the generalisability of our findings.

Third, the environmental hashtags used to retrieve Spanish videos were based on English search terms. Although we checked their associated number of plays to ensure popularity, Spanish TikTok may comprise distinct subcultures with their own prioritization and interpretation of environmental issues. Moreover, as Spanish is spoken in countries across the Global North and Global South, linking environmental discussions with their geographical origins is important to tease apart how language and geographical locations relate to specific environmental beliefs and actions. Fourth, in our analysis, we grouped different types of individual accounts (e.g. influencer accounts, activist accounts, scientist accounts) into one category which emerged to also be the largest group of social actors posting environmental content. This was because, unlike Twitter profile information, TikTok profile information is not very informative about the creator. Future studies could use surveys or detailed analyses of profile information and content to examine the variability among individual user accounts such as those of notable scientists, activists, and influencers to further examine the differences in content creation and audience engagement. Finally, we included the number of comments as a measure of engagement but did not analyse the text of these comments. An analysis of user comments can reveal whether the audience responded positively or negatively to environmental content, offering a deeper understanding of how people respond to environmental messaging.

Conclusion

We conducted a content analysis of environment related TikTok videos in English and Spanish languages to describe the message characteristics, social actors, and engagement of audiences in

terms of plays, likes, comments and shares. Environmental TikTok was found to have a diverse environmental discourse comprising topics beyond climate change, but the environmental ideologies and sentiments were not clearly expressed in most of the videos. Individual users were the most prominent and engaging social actors. Key stakeholders in environmental discussions like government and sustainability organizations received low engagement. We recommend that for higher audience engagement, social media messaging should also include novel or lesser-known environmental ideas, make explicit connections to the reasons behind adopting pro-environmental behaviors, link personal experiences with global environmental issues, and employ emotional appeals in increasing audience engagement with environmental messages.

Notes

1. We analysed all research questions (RQs) as preregistered but report the results of RQs concerning (a) associations between message characteristics and social actors and (b) association between number of hashtags and engagement briefly on the OSF project page. Detailed analysis will be reported in a separate article, which will be uploaded on the OSF project page subsequently.
2. For an in-depth description of the data collection strategy, and a discussion of the advantages and limitations of collecting data via TikTok and APIs, please see the detailed methods on OSF https://osf.io/hsnmf.

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Author contributions

Shreya Dubey: Conceptualization, Methodology, Investigation, Formal analysis, Data curation, Writing – original draft, Visualization, Project administration
Marijn Meijers: Conceptualization, Methodology, Validation, Writing – Review & Editing, Supervision, Funding acquisition
Edith Smit: Conceptualization, Methodology, Validation, Writing – Review & Editing, Supervision, Funding acquisition
Eline Smit: Conceptualization, Methodology, Validation, Writing – Review & Editing, Supervision, Funding acquisition

Disclosure statement

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Data availability

As per the current institutional policy, we cannot share the data for this study as it contains identifying information that relates to individuals. The coding sheets for all data will be shared (without

meta-data) upon reasonable requests. Shareable materials for this study (including the codebook) are available on the OSF project page https://osf.io/bewr4/.

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Appendix

Table A1. Data processing – TikTok videos excluded from analysis.

Reason for exclusion [Explanation]	Excluded videos *English*	*Spanish*
Videos documenting unrelated events [video is about unrelated events and is not linked with sustainability (e.g. Oscars, International women's day]	23	56
Hashtag bombing [creator uses environmental hashtags on unrelated content (to increase visibility of content)]	66	43
Environmentally unrelated video posted from a non-environmental personal account [the creator does not post environmental content frequently – account is mostly used for personal blogging unrelated to the environment (e.g. videos documenting daily life, videos showing photos of the creator)]	214	179
Video is posted multiple times from niche/business accounts using popular environmental hashtags [e.g. showing how to make beeswax papers, gaming videos, 5-minute crafts, mining gold from waste (video posted repeatedly; one video is already in the sample)]	48	19
Videos from environmental accounts posting non-environmental content [e.g. about Covid-19 only, about LGBTQ + issues only with no mention of the environment]	4	77
Videos from hashtags with multiple meanings [e.g. Fossil watch ad (#fossilfuels); car combustion engines (#carbon); Rainforest Café (#rainforest), non-environmental innovation (#innovation)]	83	106
Duplicate post [same video, same caption, same account]	318	239
Duplicate video [same video, different caption, same account]	0	37
Language [the video is not in English or Spanish]	623	576
Number of videos remaining	1021	1068

Table A2. Engagement with message characteristics and social actors.

Variable	Plays		Likes		Comments		Shares	
	exp(B)[a]	95% CI	exp(B)	95% CI	exp(B)	95% CI	exp(B)	95% CI
Primary topic								
Biodiversity	0.89	0.51, 1.56	1.11	0.62, 1.97	1.48	0.81, 2.70	1.02	0.54, 1.91
Energy transition	**0.38***	0.17, 0.83	**0.28*****	0.12, 0.62	0.63	0.27, 1.46	0.59	0.25, 1.43
Environmental justice	1.49	0.69, 3.21	1.33	0.61, 2.93	0.75	0.33, 1.72	0.50	0.21, 1.19
Environmental pollution	1.71	0.93, 3.17	**1.92***	1.03, 3.59	1.50	0.78, 2.89	1.10	0.56, 2.19
Extreme weather	**0.32****	0.14, 0.71	**0.39***	0.17, 0.89	**0.32****	0.13, 0.75	**0.40***	0.16, 0.99
Food-related videos	**2.05***	1.06, 3.97	**2.47****	1.26, 4.85	1.83	0.90, 3.72	1.99	0.95, 4.15
Healthy living	2.60	0.69, 9.80	3.66	0.95, 14.13	2.47	0.60, 10.18	2.74	0.62, 12.03
Sustainability	1.04	0.59, 1.83	1.25	0.70, 2.22	0.88	0.48, 1.62	0.61	0.33, 1.15
Technological innovation	1.19	0.62, 2.27	1.16	0.60, 2.25	1.97	0.98, 3.94	1.18	0.57, 2.44
Waste management	1.39	0.75, 2.59	1.63	0.86, 3.06	1.87	0.96, 3.64	1.4	0.70, 2.81
Others	1.80	0.81, 4.01	1.67	0.74, 3.77	1.00	0.43, 2.36	0.88	0.36, 2.16
Sentiment								
Anti-environmental	2.15	0.42, 11.14	1.18	0.22, 6.34	0.80	0.14, 4.65	0.39	0.10, 1.57
Pro-environmental	**0.53*****	0.40, 0.71	**0.53*****	0.39, 0.71	**0.41*****	0.30, 0.57	0.33	0.05, 2.06
Ambiguous	0.41	0.12, 1.43	0.83	0.23, 2.97	0.39	0.10, 1.46	0.39	0.10, 1.57
Ideology								
Explicit	1.54	0.94, 2.52	**2.13*****	1.29, 3.5	**2.05****	1.21, 3.46	**2.65*****	1.53, 4.58
Animal rights	**0.46*****	0.31, 0.68	**0.46*****	0.31, 0.69	**0.56****	0.36, 0.85	**0.34*****	0.22, 0.53
Deep ecology	**2.28****	1.21, 4.28	**2.69*****	1.41, 5.11	**2.38****	1.21, 4.67	**2.79*****	1.38, 5.64
Eastern traditions	0.10	0.00, 9.47	0.11	0.00, 11.35	0.07	0.00, 9.00	0.04	0.00, 6.07
Ecofeminism	1.98	0.60, 6.54	1.64	0.48, 5.55	**9.85*****	2.74, 35.43	**6.16****	1.62, 23.44
Ecological sensibility	2.32	0.70, 7.72	2.82	0.83, 9.61	1.38	0.38, 4.99	3.36	0.88, 12.86
Indigenous ideologies	2.19	0.51, 9.49	1.40	0.31, 6.25	0.59	0.12, 2.83	0.52	0.10, 2.68
Land based ethics	0.93	0.53, 1.66	1.14	0.63, 2.04	1.82*	0.99, 3.37	1.35	0.71, 2.57
Preservationism	1.41	0.86, 2.3	1.38	0.84, 2.28	0.97	0.57, 1.63	1.30	0.75, 2.25
Social ecology	1.47	0.67, 3.21	1.73	0.78, 3.82	**4.50*****	1.95, 10.36	**4.46*****	1.86, 10.65
Unrestrained instrumentalism	0.60	0.24, 1.5	0.46	0.18, 1.18	0.62	0.23, 1.65	0.75	0.27, 2.1
Audio characteristics								
Performance	1.80	0.81, 3.98	1.99	0.88, 4.47	1.73	0.74, 4.04	**2.94***	1.21, 7.14
Diegetic	**1.40***	1.05, 1.88	**1.42***	1.05, 1.91	0.91	0.67, 1.24	1.16	0.84, 1.61
Non-diegetic	1.05	0.77, 1.45	1.20	0.87, 1.66	0.98	0.69, 1.37	1.37	0.96, 1.95
Video features								
Found	1.20	0.84, 1.70	1.13	0.79, 1.62	**1.46***	1, 2.13	**1.56***	1.05, 2.31
Animated	2.19	0.91, 5.25	1.32	0.54, 3.21	1.16	0.46, 2.97	1.13	0.42, 3.00
Timelapse	1.35	0.43, 4.18	1.39	0.44, 4.41	0.48	0.14, 1.61	0.83	0.23, 2.95
Interaction	1.67	0.57, 4.89	1.28	0.43, 3.83	1.47	0.46, 4.64	1.52	0.46, 5.05
Genre								
Vlog	0.79	0.57, 1.09	0.86	0.62, 1.2	0.73	0.52, 1.04	**0.66***	0.46, 0.94
Hosted	1.71	0.92, 3.16	1.59	0.85, 2.98	**2.12***	1.1, 4.09	1.63	0.82, 3.24
Interview	0.71	0.29, 1.75	0.68	0.27, 1.70	0.93	0.35, 2.43	0.74	0.27, 2.02
Voice over pictures	1.26	0.86, 1.86	1.45	0.98, 2.16	**1.89*****	1.25, 2.87	1.50	0.97, 2.31
Text over pictures	0.73	0.46, 1.18	0.81	0.50, 1.32	0.65	0.39, 1.09	0.90	0.53, 1.53
Context								
Global	**2.69*****	1.52, 4.74	**3.19*****	1.79, 5.70	**3.04*****	1.66, 5.58	**2.63*****	1.40, 4.97
Local	1.36	0.91, 2.02	1.28	0.85, 1.92	1.17	0.76, 1.79	1.46	0.94, 2.28
Both	0.84	0.18, 3.89	0.88	0.18, 4.21	0.73	0.14, 3.75	2.09	0.38, 11.6
Not explicit	1.13	0.78, 1.63	1.17	0.80, 1.71	0.99	0.66, 1.47	1.27	0.84, 1.91
Experiences								
Personal	**1.93*****	1.28, 2.92	**1.80****	1.18, 2.75	**2.74*****	1.76, 4.26	**2.04*****	1.28, 3.23
Others'	1.85	0.87, 3.95	1.88	0.87, 4.07	**3.10****	1.38, 6.97	1.79	0.77, 4.17
Not explicit	**0.63***	0.42, 0.96	**0.60***	0.39, 0.91	1.16	0.75, 1.80	0.73	0.46, 1.16
Keying								
Funny	0.67	0.21, 2.16	0.69	0.21, 2.28	0.56	0.16, 1.95	0.61	0.16, 2.23
Ironic	2.02	0.76, 5.42	2.60	0.95, 7.11	2.69	0.94, 7.71	2.81	0.94, 8.46
Mocking	0.74	0.33, 1.64	0.60	0.26, 1.35	0.70	0.30, 1.65	0.65	0.27, 1.60

(*Continued*)

Table A2. Continued.

Variable	Plays		Likes		Comments		Shares	
	exp(B)[a]	95% CI	exp(B)	95% CI	exp(B)	95% CI	exp(B)	95% CI
Pretending	**0.16****	0.04, 0.62	**0.13*****	0.03, 0.54	**0.13****	0.03, 0.57	**0.04*****	0.01, 0.17
Communicative function								
Conative	0.80	0.65, 1.17	0.92	0.68, 1.24	1.04	0.76, 1.43	0.83	0.60, 1.15
Emotive	**3.56*****	2.04, 6.23	**4.64*****	2.62, 8.21	**4.55*****	2.50, 8.27	**3.65*****	1.95, 6.82
Metalingual	0.64	0.09, 4.80	0.90	0.12, 6.97	0.64	0.07, 5.47	1.86	0.20, 17.49
Phatic	1.02	0.31, 3.33	1.61	0.48, 5.39	1.65	0.46, 5.83	0.46	0.12, 1.74
Poetic	0.98	0.52, 1.86	0.92	0.48, 1.76	0.74	0.37, 1.47	1.07	0.52, 2.18

Note. Baseline comparison groups are as follows. Primary topic = climate change, sentiment = neutral, ideology = conservatism, audio characteristics = original, video feature = original, genre = presentation, context = unclear, experiences = unclear, keying = serious, communicative function = referential.

[a]Interpretation: an exp(B) value of 0.89 means that the number of likes for the presence of a message characteristic amounted to only 89% of the expected likes whereas a value of 2.05 means that the number of likes was 205% of the expected likes.

*p = < .05, **p = < .01, ***p < .001